

Lecture 8: Models for Human-Robot Collaboration (Part-2)

COMP3018/COMP5018: Human-Robot Interaction

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Introduction

- Today's Topics

- 1- Introduction to POMDP

- Session Learning Outcomes

By the end of today's lecture you will be able to:

- 1- Describe the different between MDP and POMDP

Partially Observable Markov Decision Process (POMDP)

- Belief State: probability distribution over states
- Belief Space: the entire probability space
- Discretize the POMDP belief space
 - Solve the resulting belief-space MDP using
 - Value iteration
 - Policy iteration
 - Any MDP solving technique

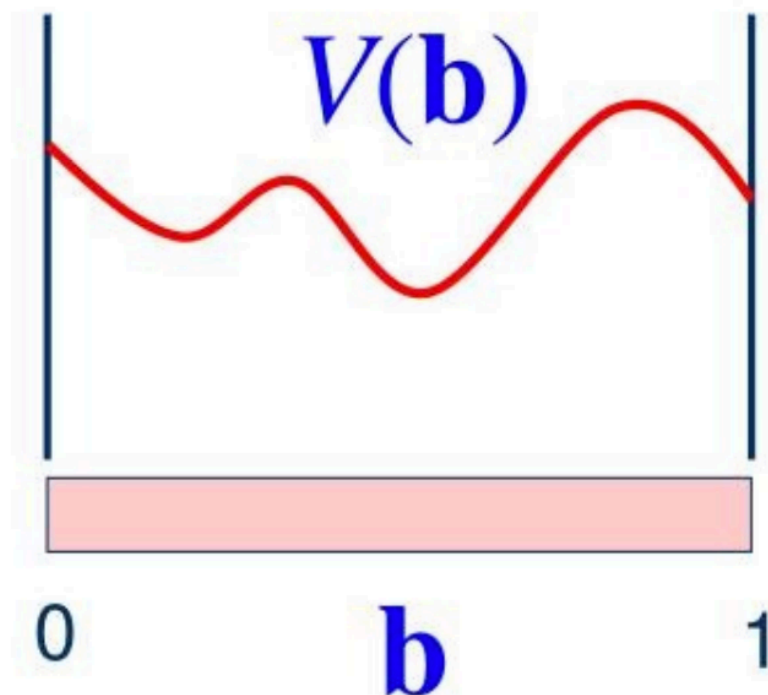
Reduce Computational Complexity

Partially Observable Markov Decision Process (POMDP)

Value Function over Belief Space

- Belief State: probability distribution over states
- Belief Space: the entire probability space

Consider a 2-state POMDP:



How to obtain the value function in belief space?

Can we use the table-based method?

We Approximate it

Partially Observable Markov Decision Process (POMDP)

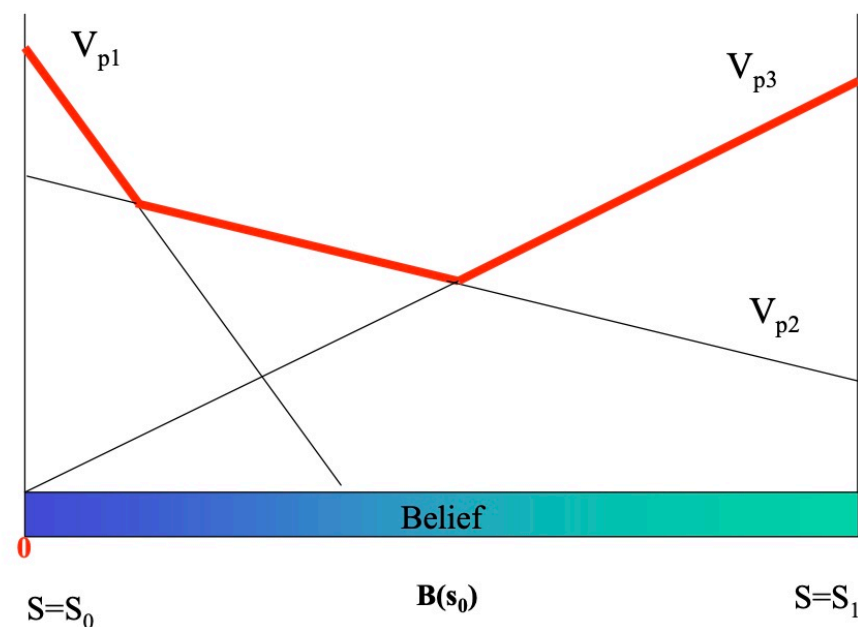
Value Function

- Corollary: A policy made up of a set of trees is **piece-wise linear** (composed of segments)
- Proof:
 - Each tree leads to a linear piece for a region of the belief space
 - Hence the value function is made up of several linear pieces.
- A **policy tree** indicates that the value function is represented hierarchically or as a tree structure. It represents the decision-making strategy over multiple time steps under uncertainty. Each node in the policy tree corresponds to a decision or action taken by the agent, and the branches from each node represent the possible observations that could result from that action.

Partially Observable Markov Decision Process (POMDP)

Optimal Value Function

- Theorem: Optimal value function $V^*(b)$ for finite horizon is **piece-wise linear and convex** in b
- Proof:
 - There are finitely many trees of finite depth
 - Each tree gives rise to a linear piece α
 - At each belief, select the highest linear piece



P is a single policy tree

- Each alpha vector corresponds to a linear segment in the value function, representing the value of being in a certain belief state and taking a specific action.

Partially Observable Markov Decision Process (POMDP)

- A Solution from Point-Based Methods:

$$V^*(b) = \max_i \left(\sum_s b(s) \cdot \alpha_i(s) \right)$$

where:

- $b(s)$ is the probability of state s in the belief state b .
 - $\alpha_i(s)$ is the value associated with state s in the i -th alpha vector.
-
- Example: let action a_1 have a value of 0 in state s_1 and 1 in state s_2 , and let action a_2 have a value of 1.5 in state s_1 and 0 in state s_2 . If our belief state is [0.75 0.25] then the value of doing action a_1 in this belief state is $0.75 \times 0 + 0.25 \times 1 = 0.25$. Similarly, action a_2 has value $0.75 \times 1.5 + 0.25 \times 0 = 1.125$.
- i.e, taking action a_2 is expected to yield a higher cumulative reward in that belief state compared to taking action a_1

Partially Observable Markov Decision Process (POMDP)

- A Solution from Reinforcement Learning:

Value Iteration

- Bellman's Equation:

$$V^*(b) = \max_a R(b, a) + \gamma \sum_{o'} \Pr(o' | b, a) V^*(b^{a,o'})$$

- Value Iteration:

– Idea: repeat

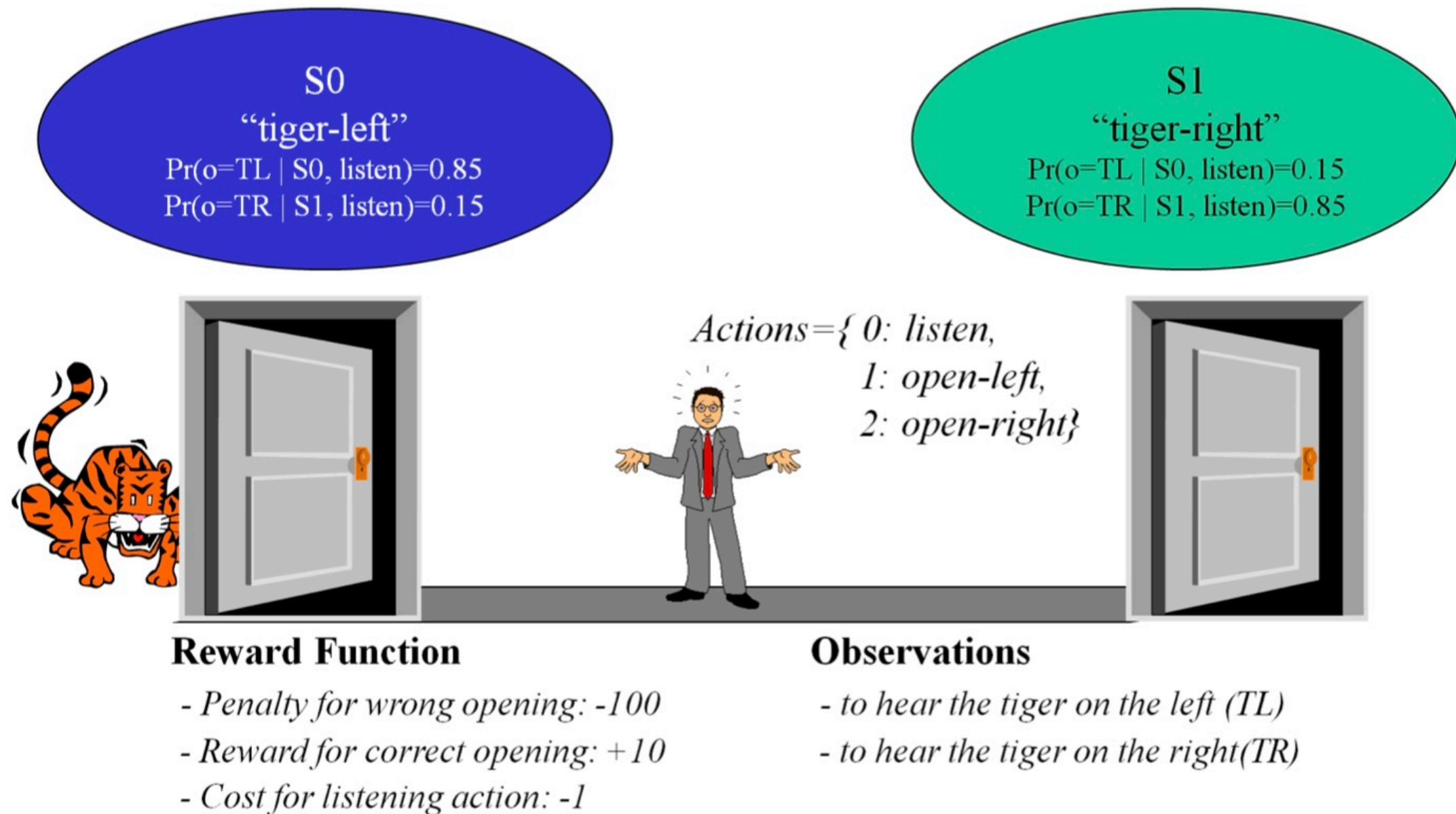
$$V^*(b) \leftarrow \max_a R(b, a) + \gamma \sum_{o'} \Pr(o' | b, a) V^*(b^{a,o'}) \quad \forall b$$

- $R(b, a)$ Immediate reward obtained from taking action a , in belief state b .
- $P(o | b, a)$ is the probability of observation o given belief state b and action a .

The best policy has the highest V^* value

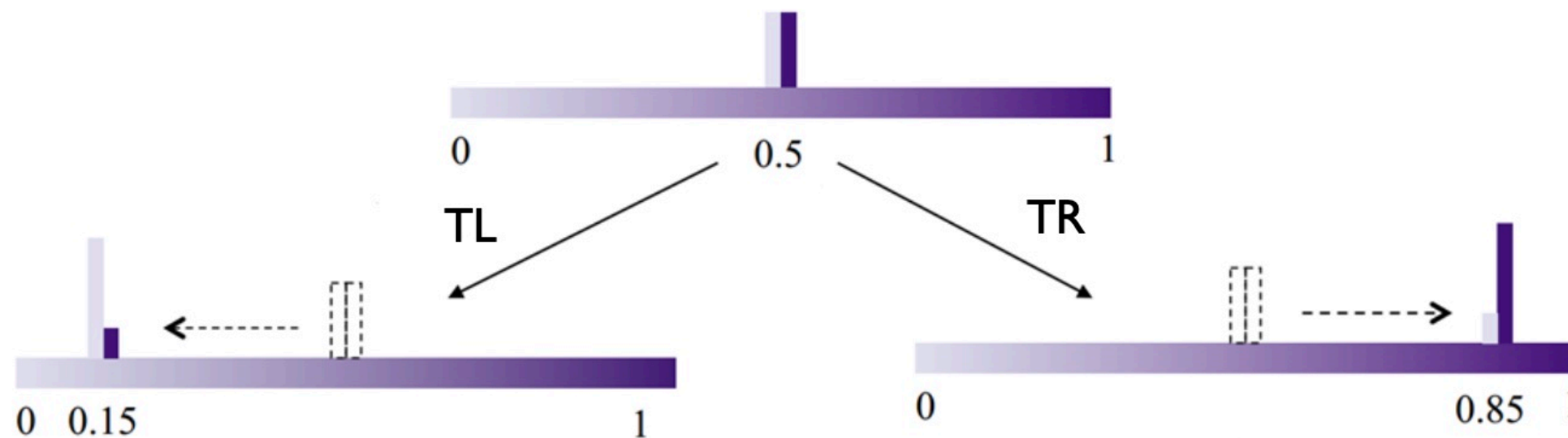
- It expresses the optimal value function recursively in terms of the immediate reward, transitions, and observations associated with the agent's actions.

Partially Observable Markov Decision Process (POMDP) - Tiger Example



Partially Observable Markov Decision Process (POMDP) - Belief State

- Probability of S0 vs S1 being true underlying state
- Initial belief state: $p(S0)=p(S1)=0.5$
- Upon listening, the belief state should change according to the Bayesian update (filtering)



Partially Observable Markov Decision Process (POMDP) - Belief State

- Each belief is a probability distribution, thus, each value in a POMDP is a function of an entire probability distribution.
- Policy π is a map from $[0,1] \rightarrow \{\text{listen, open-left, open-right}\}$
- What should the policy be?
 - Roughly: listen until sure, then open
- The solution of the POMDP is a policy prescribing which action to take in each belief state.

Solving the POMDP

Partially Observable Markov Decision

Tiger Problem (Transition Probabilities)

Prob. (LISTEN)	Tiger: left	Tiger: right
Tiger: left	1.0	0.0
Tiger: right	0.0	1.0

**Doesn't change
Tiger location**

Prob. (LEFT)	Tiger: left	Tiger: right
Tiger: left	0.5	0.5
Tiger: right	0.5	0.5

Problem reset

Prob. (RIGHT)	Tiger: left	Tiger: right
Tiger: left	0.5	0.5
Tiger: right	0.5	0.5

Partially Observable Markov Decision Process (POMDP) - Tiger Example

Tiger Problem (Observation Probabilities)

Prob. (LISTEN)	O: TL	O: TR
Tiger: left	0.85	0.15
Tiger: right	0.15	0.85

Prob. (LEFT)	O: TL	O: TR
Tiger: left	0.5	0.5
Tiger: right	0.5	0.5

Prob. (Right)	O: TL	O: TR
Tiger: left	0.5	0.5
Tiger: right	0.5	0.5

**Any observation
Without the listen action
Is uninformative**

Partially Observable Markov Decision Process (POMDP) - Tiger Example

Tiger Problem (Immediate Rewards)

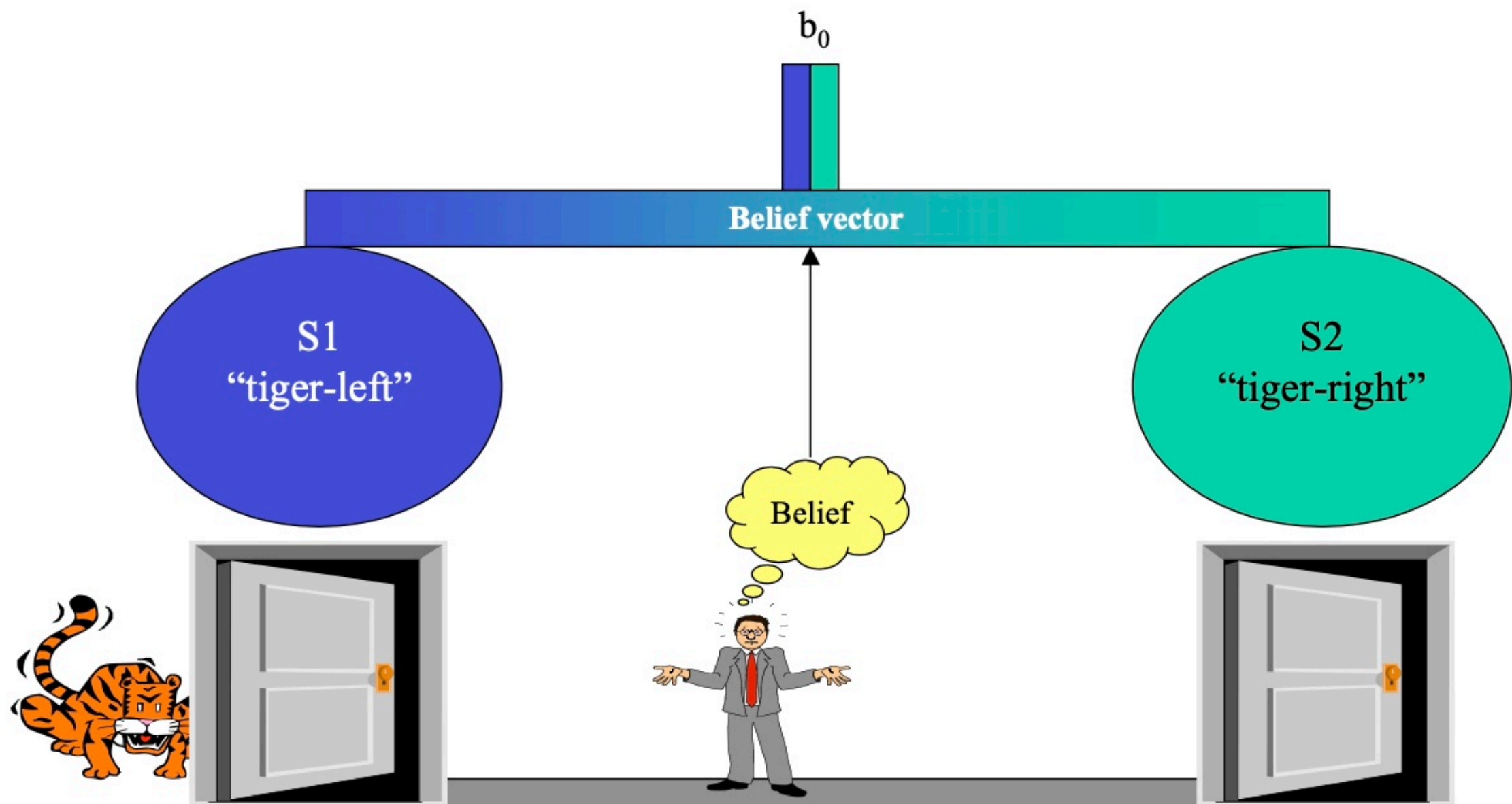
Reward (LISTEN)	
Tiger: left	-1
Tiger: right	-1

Reward (LEFT)	
Tiger: left	-100
Tiger: right	+10

Reward (RIGHT)	
Tiger: left	+10
Tiger: right	-100

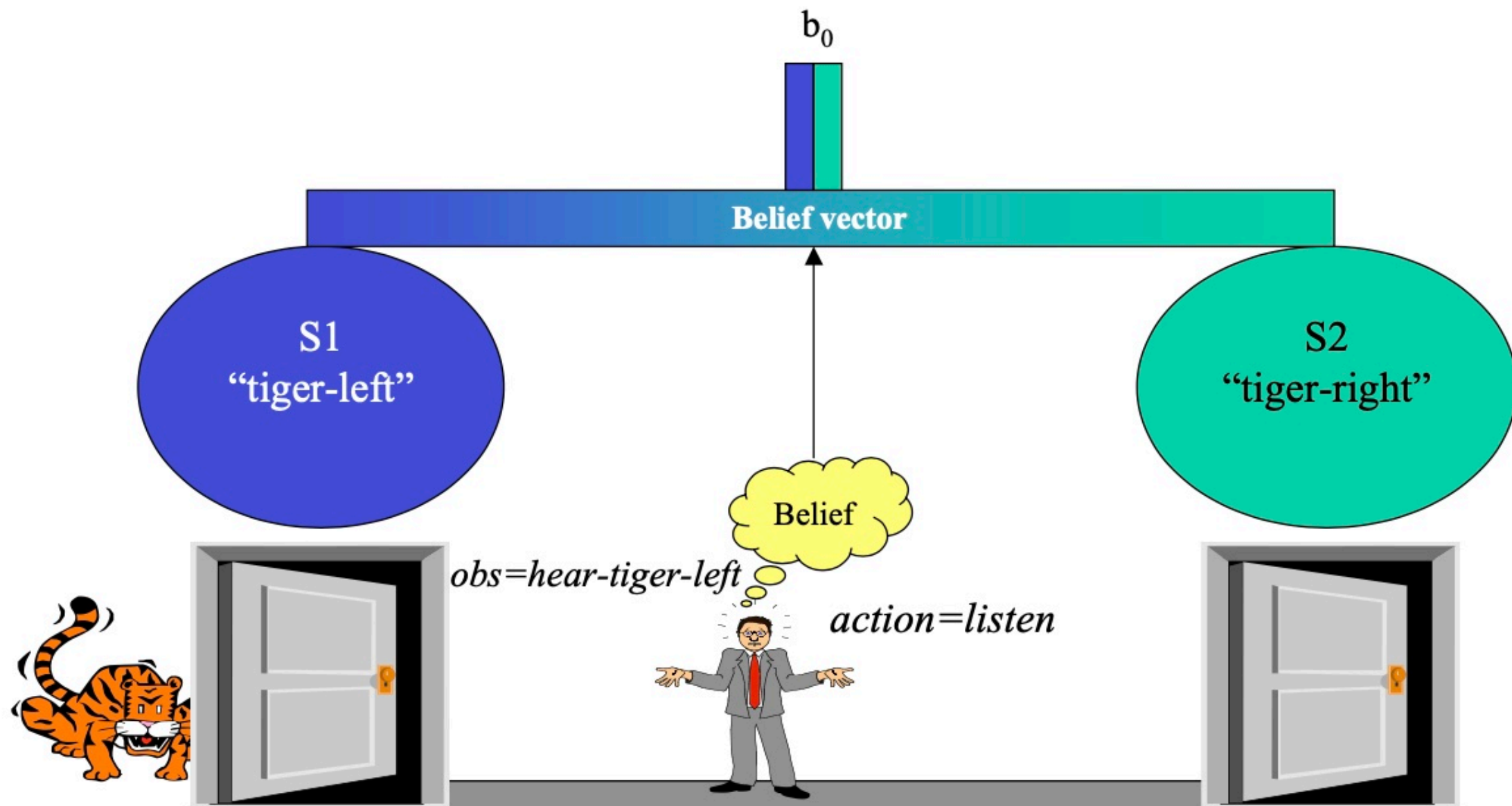
Partially Observable Markov Decision Process (POMDP) - Tiger Example

The tiger problem: State tracking



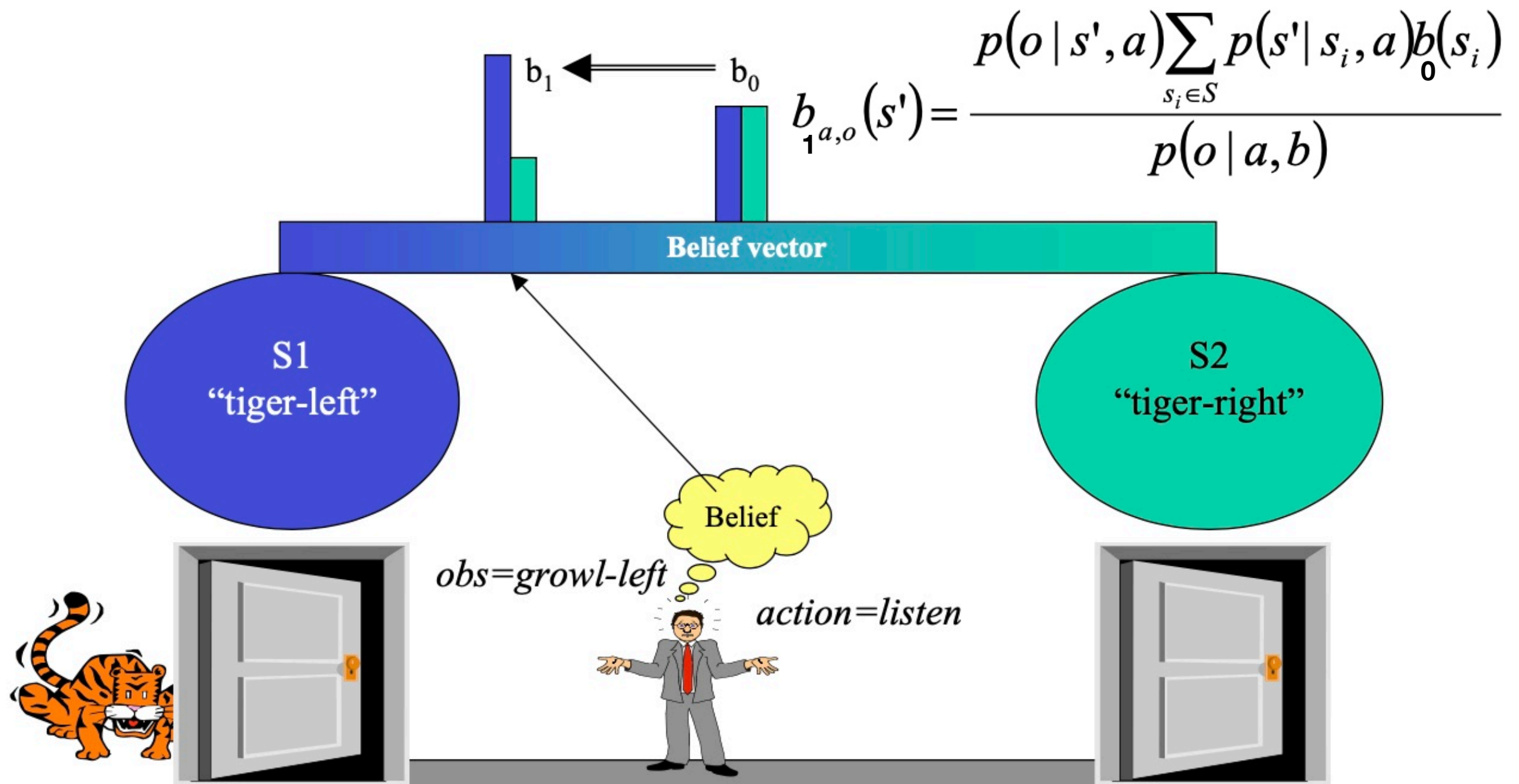
Partially Observable Markov Decision Process (POMDP) - Tiger Example

The tiger problem: State tracking



Partially Observable Markov Decision Process (POMDP) - Tiger Example

The tiger problem: State tracking



Partially Observable Markov Decision Process (POMDP) - Tiger Example

Tiger Example Optimal Policy t=1

- Optimal Policy for t=1

The approximate value function for belief states



$$\alpha^0(1)=(-100.0, 10.0)$$

$$\alpha^1(1)=(-1.0, -1.0)$$

$$\alpha^0(1)=(10.0, -100.0)$$

left

listen

right

[0.00, 0.10]

[0.10, 0.90]

[0.90, 1.00]

Beliefs

open-left

listen

open-right

Optimal policy:

Belief Space:

S1
"tiger-left"

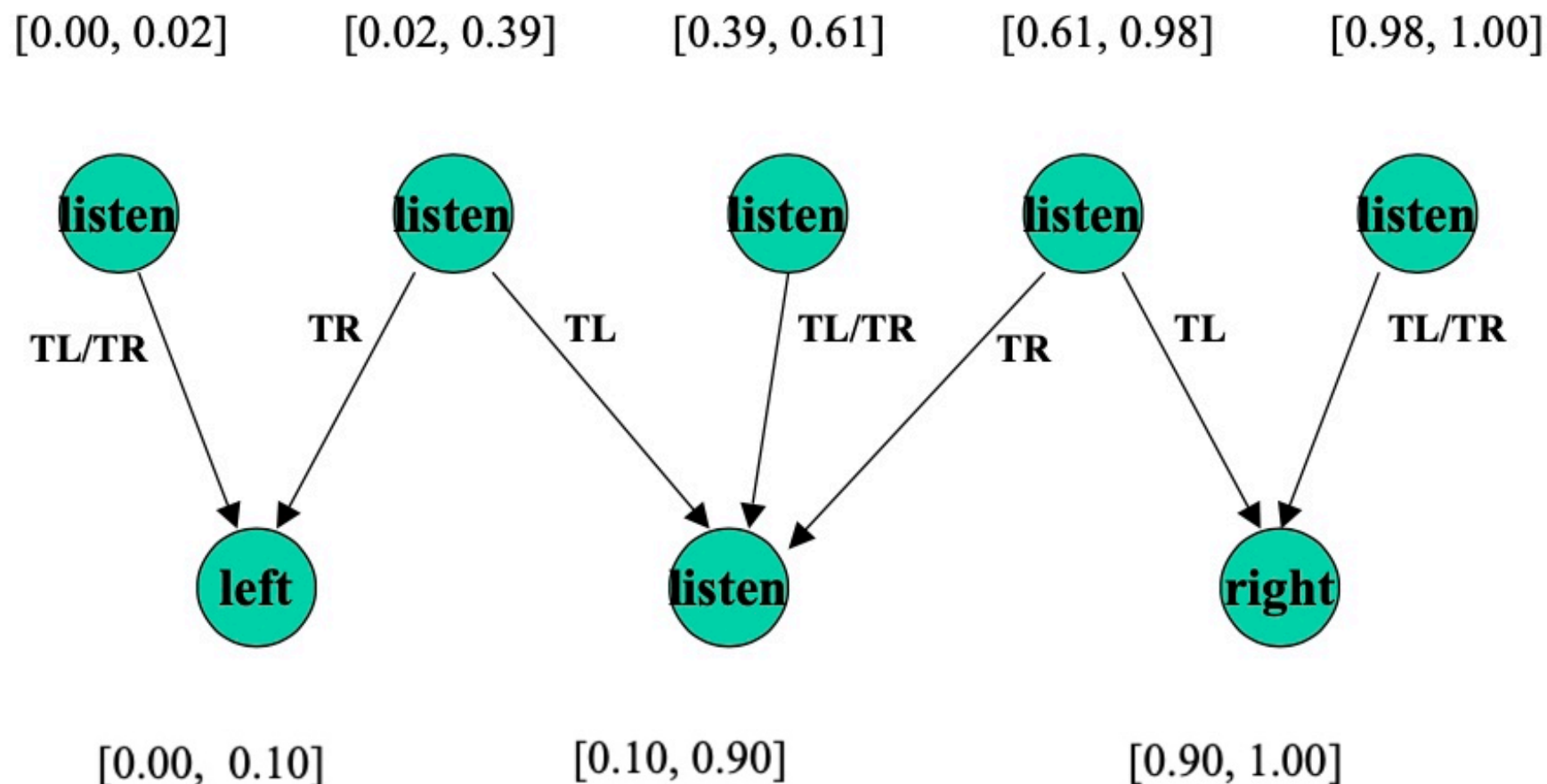
S2
"tiger-right"

This belief state indicates that the agent is moderately uncertain about the true state of the environment

Partially Observable Markov Decision Process (POMDP) - Tiger Example

Tiger Example Optimal Policy for $t=2$

- For $t=2$

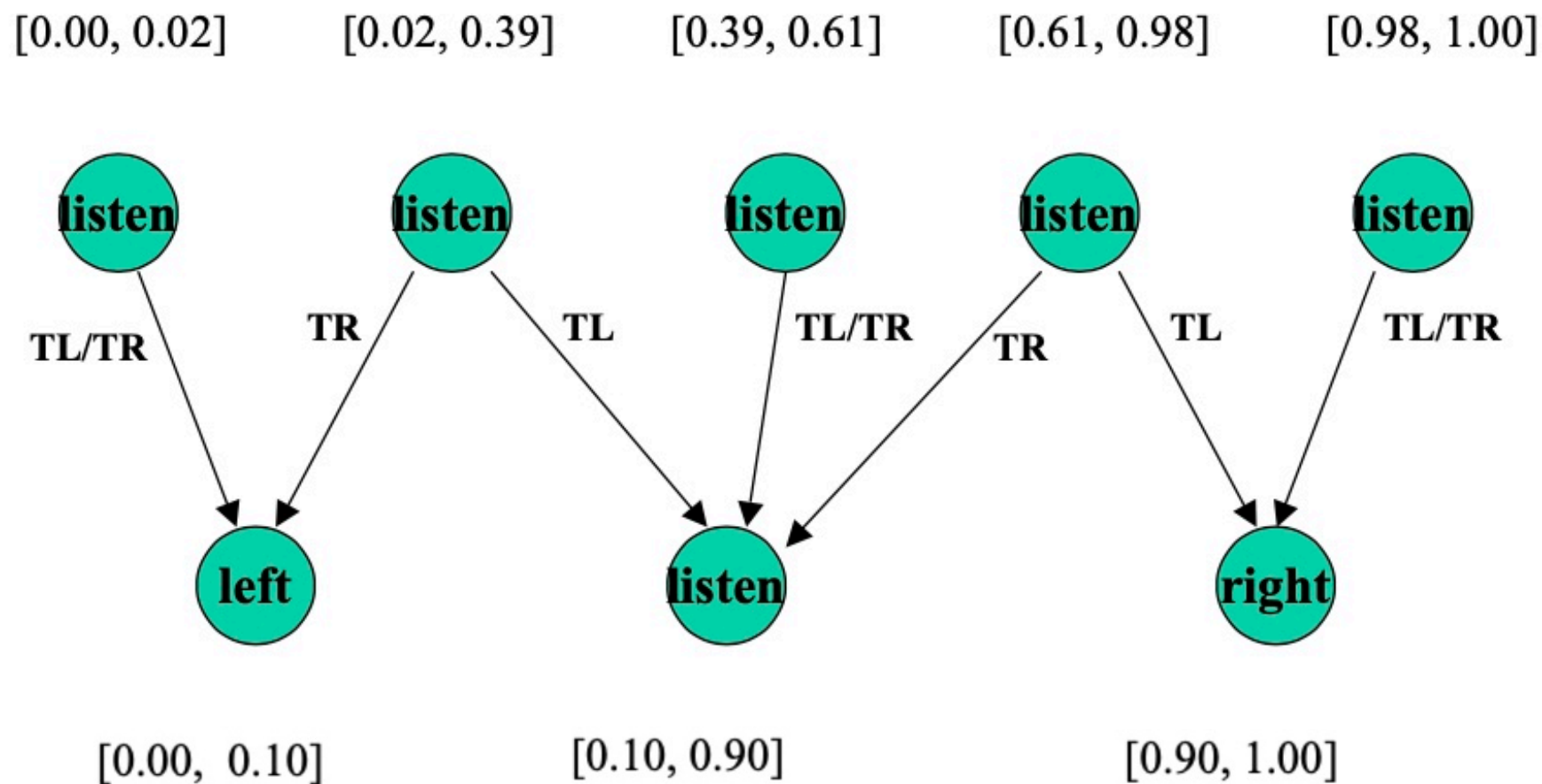


- This reflects the evolution of the decision process over time as more information is gathered or as more actions are taken.
- As time progresses to $t=2$, the agent has had more opportunities to "listen" and thus may have a more refined belief state about where the tiger is. This leads to **more branches in the decision tree**.

Partially Observable Markov Decision Process (POMDP) - Tiger Example

Tiger Example Optimal Policy for $t=2$

- For $t=2$



- The reason you see many "listen" states at $t=2$ is because each time the agent chooses to "listen" rather than open a door, it gets more information which updates its belief state about the location of the tiger.

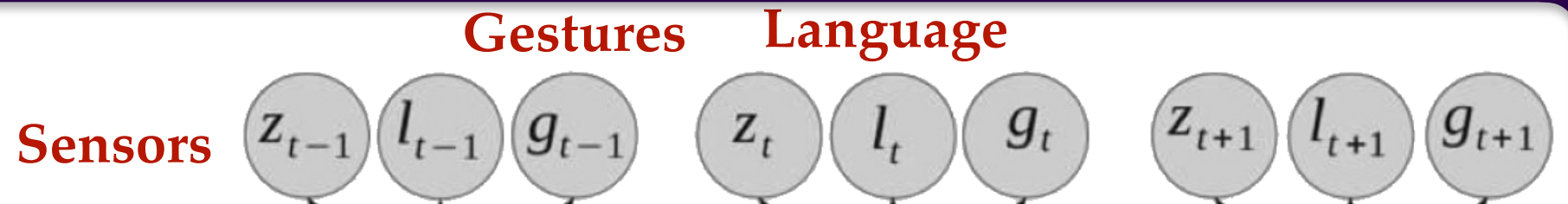
Calculate the V value using the Bellman equation and iterate at different time steps until it converges

POMDP in HRI

- Partially Observable Markov Decision Process (POMDP) for **decision-making under uncertainty**.

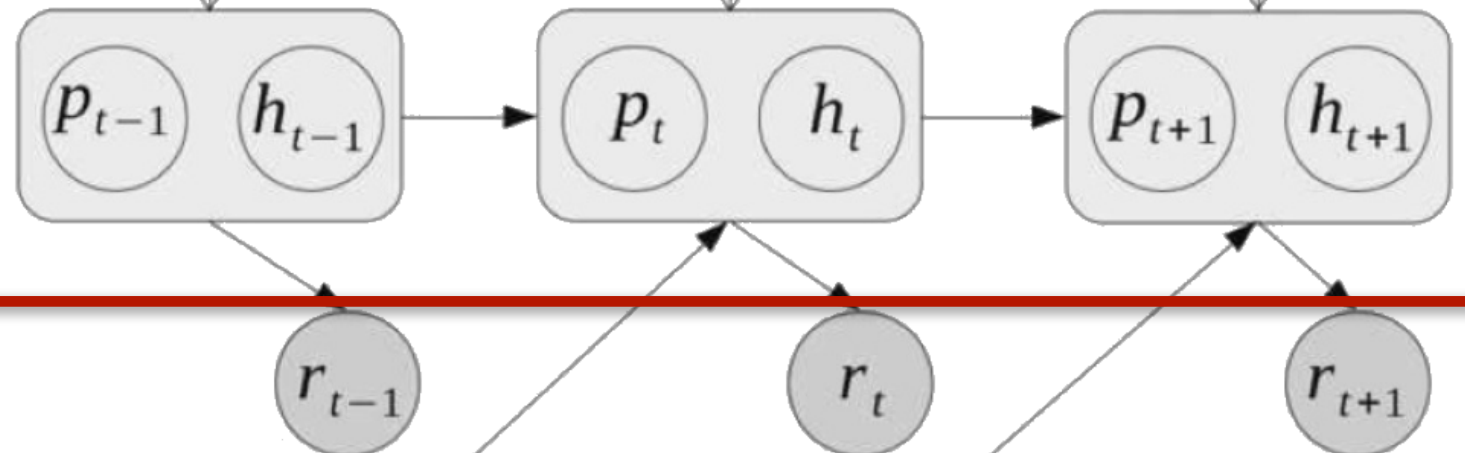
- Observations**

z_t : Sensors
 l_t : Language
 g_t : Gestures



- States**

p_t : Physical states of the world
(Objects, Robots)
 h_t : Human mental states

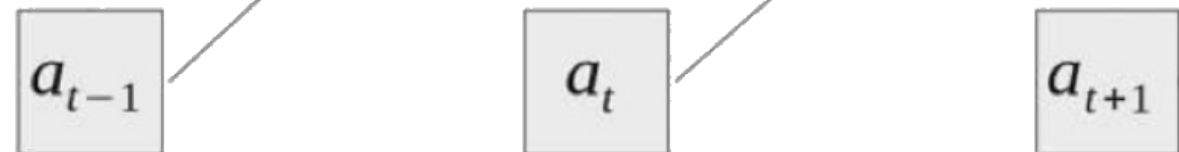


- Reward**

r_t : Do what the person wants

- Robot Actions**

a_t : Physical actions, languages, gestures

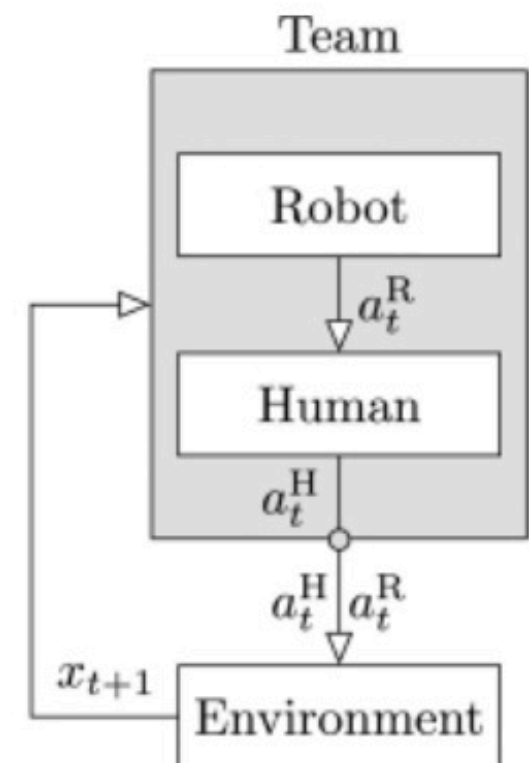
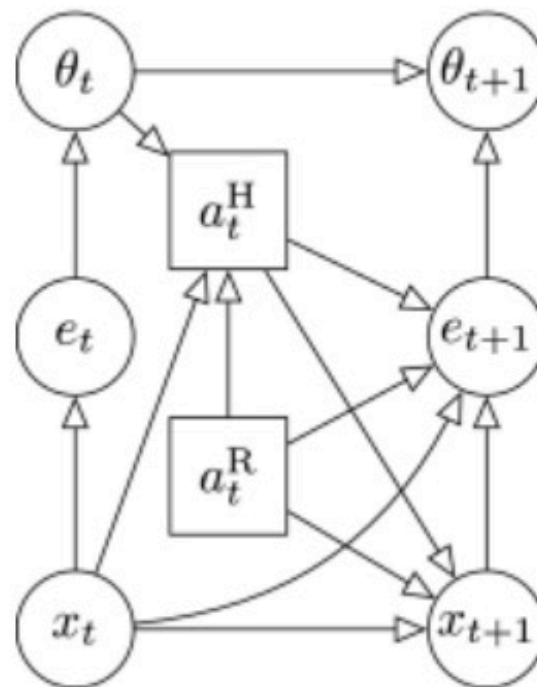


POMDP for Human-Robot Collaboration

We need a computational model that integrates trust (latent variable)
into robot decision-making

The human action a_t^H depends on the world state x_t and the human trust θ_t . The robot action a_t^R depends on the world state x_t and its belief over trust θ_t . The world state x_t transits to x_{t+1} when provided the human action a_t^H and the robot action a_t^R . The human trust θ_t transits to θ_{t+1} when provided the robot performance e_{t+1} .

The trust variable is
incorporated in the
reward function and the
Bellman equation



Trust-POMDP

Summary

- Today we discussed

- 1- Introduction to POMDP

- Next week

- 1- We will discuss Cognitive Robotics